

Exercise 4



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Macroeconomics I

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Learning By Doing: Externalities

Consider an economy described as follows:

$$u(c_t) = \frac{c_t^{1-1/\theta}}{1-1/\theta}$$

$$Y_t = K_t^\alpha (h_t L_t)^{1-\alpha}$$

$$k_{t+1} = (1 - \delta)k_t + i_t$$

$$h_t = b k_t, b > 0, L_t = L = \text{constant}, \forall t$$

Assume discount factor, β , is in $(0, 1)$.

1.

Describe how human capital, h_t , accumulates in this economy. Do agents invest in improving h_t ? Do you believe centralized, and decentralized problems, will have the same outcome in this economy?

Agents do not invest in improving human capital in this model. The human capital in this model is increased as physical capital per worker increases ($h_t = bk_t$). Meaning that workers' human capital is increased when more physical capital is allocated to them.

The social planner's does not come to the same result as decentralized problem, increase in human capital is an externality in this model. Thus, we expect that the investment in physical capital in the equilibrium would be less than the social optimum.

2.

Write and solve the social planner's problem. What is the growth rate of consumption in this economy? What is the optimal saving rate (assume a linear savings function)?

The resource constraint is as follows:

$$k_{t+1} - k_t = k_t^\alpha h_t^{1-\alpha} - \delta k_t - c_t$$

Plugging the problems assumption ($h_t = bk_t$) in it, we will reach:

$$\begin{aligned} k_{t+1} - k_t &= k_t^\alpha (bk_t)^{1-\alpha} - \delta k_t - c_t \\ &= k_t b^{1-\alpha} - \delta k_t - c_t \end{aligned}$$

The social planner wants to maximize the discounted utility subject to the resource constraint, by choosing the consumption of current and capital of the next period! So, its problem would be:

$$\max_{c_t, k_{t+1}} \sum_{t=0}^{\infty} \beta^t u(c_t) \quad \text{s.t.} \quad k_{t+1} - k_t = k_t b^{1-\alpha} - \delta k_t - c_t$$

Writing the Lagrangian, we have:

$$\mathcal{L} = \sum_{t=0}^{\infty} \beta^t u(c_t) + \sum_{t=0}^{\infty} \mu_t \left[(1 + b^{1-\alpha} - \delta)k_t - c_t - k_{t+1} \right]$$

Taking the FOC, we get:

$$\frac{\partial \mathcal{L}}{\partial c_t} = \beta^t u'(c_t) - \mu_t = 0 \quad (*)$$

$$\frac{\partial \mathcal{L}}{\partial k_{t+1}} = -\mu_t + \mu_{t+1}(1 + b^{1-\alpha} - \delta) = 0 \quad (**)$$

Plugging μ_t & μ_{t+1} from $\star$ into $\star\star$, we will reach the Euler equation:

$$\begin{aligned} -\mu_t + \mu_{t+1}(1 + b^{1-\alpha} - \delta) &= 0 \\ -\beta^t u'(c_t) + \beta^{t+1} u'(c_{t+1})(1 + b^{1-\alpha} - \delta) &= 0 \\ \beta^{t+1} u'(c_{t+1})(1 + b^{1-\alpha} - \delta) &= \beta^t u'(c_t) \\ \implies \frac{u'(c_{t+1})}{u'(c_t)} &= \frac{1}{\beta(1 + b^{1-\alpha} - \delta)} \quad \checkmark \end{aligned}$$

Previously we had:

$$u(c_t) = \frac{c_t^{1-1/\theta}}{1-1/\theta} \implies u'(c_t) = c_t^{-1/\theta}$$

Plugging into the Euler equation, we get:

$$\frac{c_{t+1}}{c_t} = [\beta(1 + b^{1-\alpha} - \delta)]^\theta \quad (\star\star\star)$$

Assuming a linear saving function with rate s , we have:

$$c_t = (1 - s)(1 + b^{1-\alpha} - \delta)k_t \quad (1)$$

$$c_{t+1} = (1 - s)(1 + b^{1-\alpha} - \delta)k_{t+1} \quad (2)$$

Also we have:

$$k_{t+1} = s(k_t b^{1-\alpha} - \delta k_t) = s(b^{1-\alpha} - \delta)k_t \quad (3)$$

Plugging (1) and (2) into the $\star\star\star$, we get:

$$\begin{aligned} \frac{c_{t+1}}{c_t} &= [\beta(1 + b^{1-\alpha} - \delta)]^\theta \\ \frac{\cancel{(1-s)(1+b^{1-\alpha}-\delta)} k_{t+1}}{\cancel{(1-s)(1+b^{1-\alpha}-\delta)} k_t} &= [\beta(1 + b^{1-\alpha} - \delta)]^\theta \end{aligned}$$

Plugging (3) into the last equation and solving for s , we finally reach the optimal saving rate, which is:

$$s = \beta^\theta (1 + b^{1-\alpha} - \delta)^{\theta-1}$$

3.

Show that the household budget constraint can be written as:

$$c_t + k_{t+1} + b_{t+1} = w_t h_t + (1 + r_t - \delta)k_t + (1 + \underbrace{R_t}_{\text{or } R_{t-1}})b_t$$

where $b + t$ is a risk-free bond with return R_t (determined endogenously). What is the arbitrage condition between risk-free bonds and capital? Use the FOCs of the household to find Euler equation. Using the firm's profit maximizing behavior, what is the equilibrium interest rate for physical capital and the wage rate? What is the growth rate of consumption?

The income sources for households are effective wages $w_t h_t$, return on capital $r_t k_t$ and return on bonds $R_t b_t$. In each period, the household can either consume the income c_t , or invest in the physical capital i_t^k , or buy the government bonds i_t^b . So we can write the household budget constraint as:

$$\begin{aligned} c_t + i_t^k + i_t^b &= w_t h_t + r_t k_t + R_t b_t \\ c_t + (k_{t+1} - k_t + \delta k_t) + i_t^b &= w_t h_t + r_t k_t + R_t b_t && \text{(Using } i_t^k = k_{t+1} - k_t + \delta k_t) \\ c_t + (k_{t+1} - k_t + \delta k_t) + (b_{t+1} - b_t) &= w_t h_t + r_t k_t + R_t b_t && \text{(Using } i_t^b = b_{t+1} - b_t) \\ c_t + k_{t+1} + b_{t+1} &= w_t h_t + (1 + r_t - \delta)k_t + (1 + R_t)b_t \quad \checkmark && (*) \end{aligned}$$

In order to not having arbitrage, the following equation of return of capital and bond must hold. Otherwise the agents will prefer one type of investment to the other.

$$R_t = r_t - \delta$$

Plugging into (*), we get:

$$c_t + k_{t+1} + b_{t+1} = w_t h_t + (1 + R_t)(k_t + b_t) = w_t h_t + (1 + (r_t - \delta))(k_t + b_t)$$

The no-arbitrage condition implies that there is no difference in investing in capital or bonds, So basically we can define asset in period t as: $a_t = k_t + b_t$. Plugging into the previous equation, we get:

$$c_t + k_{t+1} + b_{t+1} = w_t h_t + (1 + R_t)(a_t) = w_t h_t + (1 + (r_t - \delta))(a_t)$$

The household is choosing its consumption in the current period and its asset in the next period to maximize its aggregate utility subject to the budget constraint. So its problem would be:

$$\max_{c_t, a_{t+1}} \sum_{t=0}^{\infty} \beta^t u(c_t) \quad \text{s.t.} \quad c_t + k_{t+1} + b_{t+1} = w_t h_t + (1 + R_t)(a_t)$$

So the lagrangian for this problem is:

$$\mathcal{L} = \sum_{t=0}^{\infty} \beta^t u(c_t) + \sum_{t=0}^{\infty} \mu_t \left[w_t h_t + (1 + R_t) a_t - c_t - a_{t+1} \right]$$

Taking FOC, we get:

$$\frac{\partial \mathcal{L}}{\partial c_t} = \beta^t u'(c_t) - \mu_t = 0 \quad (1)$$

$$\frac{\partial \mathcal{L}}{\partial a_{t+1}} = -\mu_t + (1 + R_t) \mu_{t+1} = 0 \quad (2)$$

Plugging μ_t & μ_{t+1} from (1) into (2), we will reach the Euler equation:

$$\begin{aligned} -\mu_t + (1 + R_t) \mu_{t+1} &= 0 \\ -\beta^t u'(c_t) + (1 + R_t) \beta^{t+1} u'(c_{t+1}) &= 0 \\ (1 + R_{t+1}) \beta^{t+1} u'(c_t) &= \beta^t u'(c_t) \\ \implies \frac{u'(c_{t+1})}{u'(c_t)} &= \frac{1}{\beta(1 + R_t)} \quad \checkmark \end{aligned}$$

Plugging $u'(c) = c^{-1/\theta}$ into the Euler equation:

$$\frac{c_{t+1}}{c_t} = (\beta(R_t + 1))^\theta$$

The firm problem is to maximize the profit, choosing capital and labor:

$$\max_{K,L} \Pi = K_t^\alpha (h_t L_t)^{1-\alpha} - w_t h_t L_t - r_t K_t$$

Taking the first order condition we have:

$$\begin{aligned} \frac{\partial \Pi}{\partial K} &= \alpha K_t^{\alpha-1} (h_t L_t)^{1-\alpha} - r_t = 0 \\ \implies r_t &= \alpha K_t^{\alpha-1} (h_t L_t)^{1-\alpha} \\ &= \alpha K_t^{\alpha-1} L_t^{-(\alpha-1)} h_t^{1-\alpha} \\ &= \alpha k_t^{\alpha-1} h_t^{1-\alpha} \\ \frac{\partial \Pi}{\partial L} &= (1-\alpha) K_t^\alpha h_t^{1-\alpha} L_t^{-\alpha} - w_t h_t = 0 \\ \implies w_t &= (1-\alpha) K_t^\alpha h_t^{-\alpha} L_t^{-\alpha} \\ &= (1-\alpha) k_t^\alpha h_t^{-\alpha} \end{aligned}$$

From the previous part, we have the Euler equation, substituting R_t, r_t into it, we get the consumption growth:

$$\begin{aligned} \frac{c_{t+1}}{c_t} &= [\beta(1 + R_t)]^\theta && \text{(Using } R_t = r_t - \delta) \\ \frac{c_{t+1}}{c_t} &= [\beta(1 + (r_t - \delta))]^\theta && \text{(Using } r_t = \alpha k_t^{\alpha-1} h_t^{1-\alpha}) \\ \frac{c_{t+1}}{c_t} &= [\beta(1 + (\alpha k_t^{\alpha-1} h_t^{1-\alpha} - \delta))]^\theta && \text{(Using } h_t = b k_t) \\ \frac{c_{t+1}}{c_t} &= [\beta(1 + (\alpha b k_t - \delta))]^\theta \end{aligned}$$

4.

Compare the growth rate in centralized and decentralized problems. How are they different, and why?

The growth rate of consumption in the central planner problem is as follows:

$$\frac{c_{t+1}}{c_t} = [\beta(1 + b^{1-\alpha} - \delta)]^\theta$$

The growth rate of consumption in the decentralized problem is as follows:

$$\frac{c_{t+1}}{c_t} = [\beta(1 + \alpha b^{1-\alpha} - \delta)]^\theta$$

We know that $\alpha < 1$, implying that the growth rate in the central planner problem is higher than the decentralized problem.

It is due to positive externality of learning by doing, which is not considered in the decentralized problem but is considered in the central planner's problem.

5.

Suppose the government subsidizes the private cost of capital for firms. In other words, firms must pay $(1 - \tau)r_t$ instead of r_t for each unit of capital employed. The government pays for this subsidy via lump sum taxes, T , on individuals. Solve the decentralized problem in this economy.

Rerwiting the budget constraint and implementing the tax, we have:

$$\frac{c_{t+1}}{c_t} = c_t + k_{t+1} + b_{t+1} = w_t h_t + (1 + r_t - \delta)k_t + (1 + R_t)b_t - T_t$$

The FOCs of the centralized problem and no-arbitrage will not change due to the lump sum tax.

So the growth rate of consumption will still be the same as the last part:

$$\frac{c_{t+1}}{c_t} = [\beta(1 + b^{1-\alpha} - \delta)]^\theta \quad (1)$$

On the other hand, firm's problem has changed due to the subsidy on the private cost of capital.

Rewriting the firm's problem:

$$\max_{K,L} \Pi = K_t^\alpha (h_t L_t)^{1-\alpha} - w_t h_t L_t - (1 - \tau)r_t K_t$$

Taking the first order condition we have:

$$\begin{aligned} \frac{\partial \Pi}{\partial K} &= \alpha K_t^{\alpha-1} (h_t L_t)^{1-\alpha} - (1 - \tau)r_t = 0 \\ \implies r_t &= \frac{\alpha}{(1 - \tau)} K_t^{\alpha-1} (h_t L_t)^{1-\alpha} \\ &= \frac{\alpha}{(1 - \tau)} K_t^{\alpha-1} L_t^{-(\alpha-1)} h_t^{1-\alpha} \\ &= \frac{\alpha}{1 - \tau} k_t^{\alpha-1} h_t^{1-\alpha} \\ \frac{\partial \Pi}{\partial L} &= (1 - \alpha) K_t^\alpha h_t^{1-\alpha} L_t^{-\alpha} - w_t h_t = 0 \\ \implies w_t &= (1 - \alpha) K_t^\alpha h_t^{1-\alpha} L_t^{-\alpha} \\ &= (1 - \alpha) k_t^\alpha h_t^{-\alpha} \end{aligned}$$

From the previous part, we have the Euler equation, substituting R_t, r_t, b into it; we get the consumption growth:

$$\frac{c_{t+1}}{c_t} = [\beta(1 + R_t)]^\theta \quad (\text{Using } R_t = r_t - \delta)$$

$$\frac{c_{t+1}}{c_t} = [\beta(1 + (r_t - \delta))]^\theta \quad (\text{Using } r_t = \frac{\alpha}{1 - \tau} k_t^{\alpha-1} h_t^{1-\alpha})$$

$$\frac{c_{t+1}}{c_t} = [\beta(1 + \frac{\alpha}{1 - \tau} k_t^{\alpha-1} h_t^{1-\alpha} - \delta)]^\theta \quad (\text{Using } h_t = b k_t)$$

$$\frac{c_{t+1}}{c_t} = [\beta(1 + \frac{\alpha}{1 - \tau} b^{1-\alpha} - \delta)]^\theta$$

6.

What is the optimal level of τ (for which the centralized and decentralized problems coincide)? Why does the government want to subsidize investment in this model?

For central planner's problem growth rate with subsidy to coincide with the decentralized problem, we should have:

$$\begin{aligned} [\beta(1 + b^{1-\alpha} - \delta)]^\theta &= [\beta(1 + \frac{\alpha}{1-\tau}b^{1-\alpha} - \delta)]^\theta \\ \implies 1 &= \frac{\alpha}{1-\tau} \\ \implies 1 - \tau &= \alpha \\ \implies \tau &= \alpha + 1 \end{aligned}$$

The government is willing to subsidize investment because firms do not invest in physical capital at the social optimal level. The social optimal level in this model is higher than the firm's optimal level because of the learning by doing positive externality. So the government is willing to increase the investment level to the social optimal level by giving subsidy on the investment.